# Activity 4.2: Multimodal Signal Fusion

## 1. Activity Overview & Objective

In this activity, signals generated by the **Vision Module** and **Audio Module** are synchronized and aggregated into a unified data representation. Visual indicators such as presence duration, movement patterns, and inactivity periods are aligned with corresponding audio indicators—including noise levels and speech activity—using timestamp-based synchronization.

> **Core Engineering Objective:** The fusion process ensures that both modalities contribute proportionally to engagement understanding. By correlating visual attention cues with audio interaction levels, the system captures a far more accurate picture of classroom dynamics than single-modality analysis, forming a structured, extensible input for higher-level reasoning and scoring mechanisms.

## 2. Timestamp-Based Synchronization & Extensible Data Schema

Because video frames and audio buffers are processed at distinct sampling rates, raw unimodal events cannot be compared directly without temporal alignment. The fusion pipeline aggregates asynchronous stream events into discrete, timestamp-aligned time windows and serializes them into a standardized dictionary structure (`session_data`) designed to support future modalities (such as digital interaction logs or wearable telemetry).

### 2.1 Synchronized Modality Indicators

| Modality Stream | Aligned Indicators | Synchronization & Aggregation Logic | Proportional Contribution |
|---|---|---|---|
| **Vision Module** | Presence duration, movement patterns, inactivity periods | Frame-level bounding box persistence and optical/pose motion deltas aggregated over sliding UTC timestamp windows. | Quantifies physical attendance stability, visual focus continuity, and motor restlessness or lethargy. |
| **Audio Module** | Ambient noise levels (dB), speech activity (VAD) | Audio energy envelopes and voice activity segments resampled and bucketed to match the exact visual window timestamps. | Differentiates productive verbal participation and active lecturing from disruptive noise or passive silence. |
| **Fused Representation** | Unified `session_data` payload | Cross-modal correlation joining visual attention cues with concurrent acoustic interaction levels into an extensible schema. | Feeds deterministic scoring engines and LLM-based teaching assistants with holistic classroom context. |

## 3. Downstream AI Suggestion Engine Implementation

Once the synchronized multimodal representation (`session_data`) is constructed, it is passed to higher-level reasoning and scoring services. As implemented in `backend/core/ai_suggestions.py`, the `AISuggestionEngine` consumes the fused session payload and queries the `llama3-8b-8192` model to synthesize real-time teaching suggestions, backed by structured response parsing and deterministic exception handling.

### 3.1 Reference Implementation ( `backend/core/ai_suggestions.py` )

```
# File: backend/core/ai_suggestions.py

class AISuggestionEngine:
    def generate_suggestions(self, session_data: dict):
        """
        Consumes the timestamp-synchronized multimodal session_data representation
        (presence duration, movement patterns, inactivity periods, noise levels,
        and speech activity) to generate actionable teaching suggestions.
        """
        prompt = self._build_fusion_prompt(session_data)

        try:
            completion = self.client.chat.completions.create(
                messages=[
                    {"role": "system", "content": "You are an expert AI teaching
assistant."},
                    {"role": "user", "content": prompt}
                ],
                model="llama3-8b-8192",
                temperature=0.7,
                max_tokens=200
            )

            response = completion.choices[0].message.content
            return self._parse_response(response)

        except Exception as e:
            print(f"AI Suggestion Error: {e}")
            return self._fallback_suggestions(session_data)
```

### 3.2 Architectural Highlights

- **Proportional Cross-Modal Correlation:** Aligning presence duration and inactivity periods with speech activity prevents single-modality bias, enabling the engine to distinguish quiet concentration from disengagement.
- **Extensible Data Structure:** Encapsulating fused indicators inside the unified `session_data: dict` interface decouples upstream sensor synchronization from downstream reasoning, allowing new features or modalities to be integrated without breaking existing consumers.
- **Resilient Inference Pipeline:** Utilizing `llama3-8b-8192` (`temperature=0.7`, `max_tokens=200`) paired with `self._parse_response(response)` and `self._fallback_suggestions(session_data)` ensures reliable pedagogical feedback even under network or API failures.